Run report: factor-base hyperparameter sweep

Run started 2026-10-14 18:04 UTC (`data/run_start_utc.txt`). The command was `cargo run --release --example fb_hyperparam_sweep -- research/fb_hyperparam_sweep_20261014/data` at source commit `31199a70a`, with the protocol frozen in that commit (`PROTOCOL.md`). The master seed was `0xFB5EE920261014`. Host: 4 vCPU Intel Xeon @ 2.10 GHz, x86_64, shared, L0 isolation, all work on 1 thread. Total wall time 291 s. Raw records are in `../../data/sweep.json` and `../../data/console.log`.

What ran

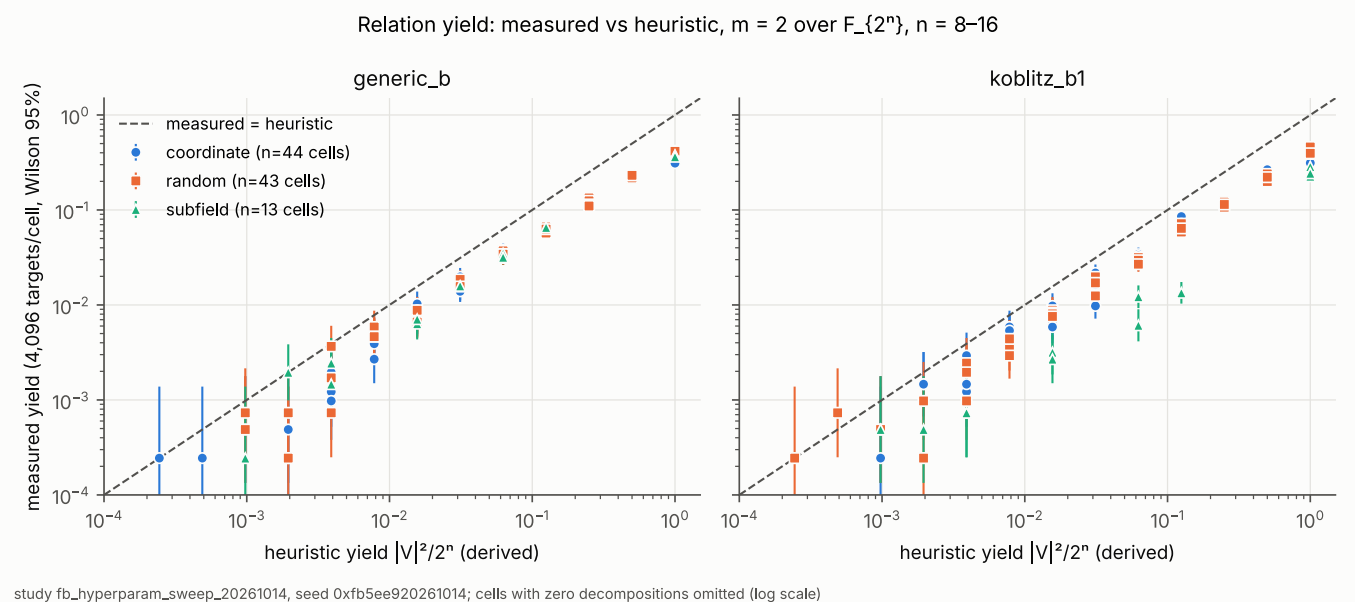
Binary arm. 208 cells: $n = 8…16$, $n' = 2…\min(6, n-1)$, basis family coordinate / random / subfield ($n' \mid n$), curve class generic b (4 curves) / Koblitz $b = 1$. Per cell: 4,096 yield targets, 256 brute-force cross-checks, 64 planted targets, 16 non-decomposable targets for d_{ff} and D^*, 4 decomposable calibration targets, $d_{\max} = 6$. 208 of 208 cells completed.

Prime arm. 20-, 24- and 28-bit $a = -3$ curves with 3 secrets each. Each secret gets one rho solve, then `ic_solve` with $m = 2$ (FB 64 … 16,384) and $m = 3$ (FB 64 … 4,096). That is 90 solves requested and 81 completed with the correct logarithm. 9 returned no solve: $m = 3$ at FB 1,024 and 4,096 (20 bits), and at FB 4,096 (24 bits). `ic_solve` does not distinguish an exhausted relation budget from a linear-algebra failure, so the reason is recorded as "relation budget exhausted or LA failed".

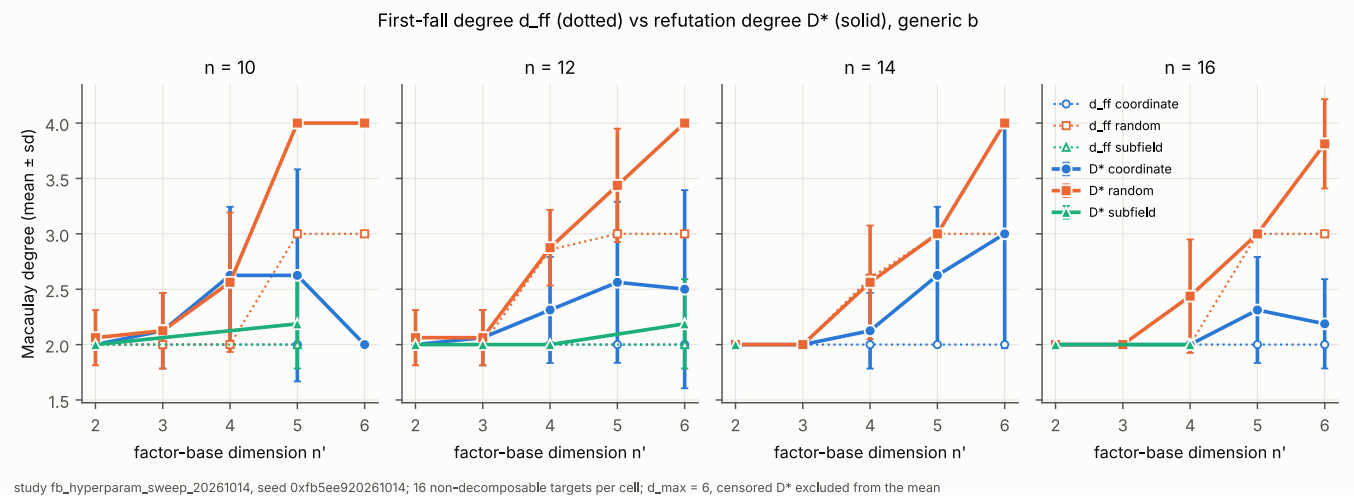
Controls (measured)

Planted targets found by root finding	13,312 / 13,312
Planted targets found by brute force (first 16 per cell)	3,328 / 3,328
Root finding = brute force on cross-checked targets	53,248 / 53,248
Decomposable calibration targets that refuted	0 / 832
Non-decomposable targets refuted by $d_{\max}$	3,264 / 3,264

Measured values

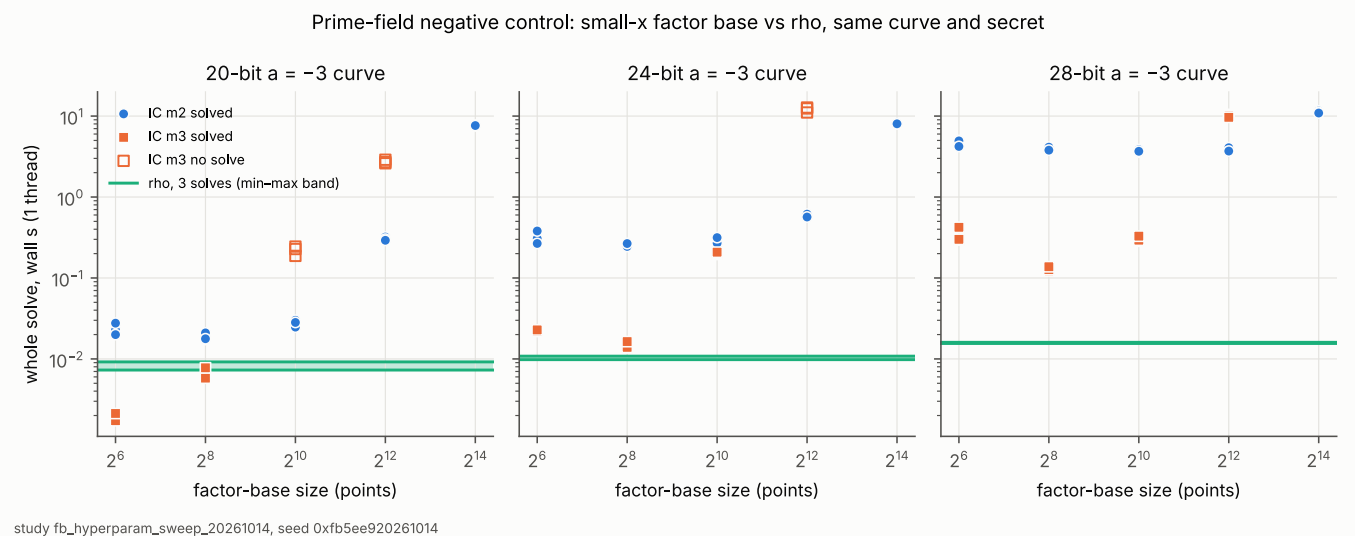


Yield per cell (4,096 targets, Wilson 95%) against $|V|^2 \ /2^n$, which is a derived reference.



Mean ± sd of d_{ff} and D^* over 16 non-decomposable targets per cell, for generic b and $n \in \{10, 12, 14, 16\}$. Every cell is in `sweep.json`.

bits	method	FB size	solved	wall ms (min–max of 3)
20	rho (negation, 1 thread)	—	3/3	7.3–9.2
20	IC m=2	64	3/3	20.0–27.7
20	IC m=2	256	3/3	17.4–21.0
20	IC m=2	1024	3/3	24.7–30.0
20	IC m=2	4096	3/3	292.0–320.3
20	IC m=2	16384	3/3	7578.3–7726.9
20	IC m=3	64	3/3	1.7–2.1
20	IC m=3	256	3/3	5.8–7.8
20	IC m=3	1024	0/3	187.4–244.4
20	IC m=3	4096	0/3	2610.1–2889.8
24	rho (negation, 1 thread)	—	3/3	9.8–10.8
24	IC m=2	64	3/3	266.9–380.5
24	IC m=2	256	3/3	245.2–266.3
24	IC m=2	1024	3/3	257.9–315.4
24	IC m=2	4096	3/3	566.3–614.8
24	IC m=2	16384	3/3	8022.9–8050.4
24	IC m=3	64	3/3	21.8–22.8
24	IC m=3	256	3/3	13.7–16.4
24	IC m=3	1024	3/3	195.5–208.2
24	IC m=3	4096	0/3	11071.4–12757.4
28	rho (negation, 1 thread)	—	3/3	15.5–16.0
28	IC m=2	64	3/3	4216.7–4994.7
28	IC m=2	256	3/3	3792.1–4131.1
28	IC m=2	1024	3/3	3588.3–3849.9
28	IC m=2	4096	3/3	3694.6–4066.5
28	IC m=2	16384	3/3	10903.4–11590.1
28	IC m=3	64	3/3	294.1–422.5
28	IC m=3	256	3/3	122.6–137.8
28	IC m=3	1024	3/3	284.5–328.5
28	IC m=3	4096	3/3	9643.6–10475.4



Every number above is measured, except the heuristic reference line. The derived cost model and the protocol's decision rules are in the results report (`../../README.md`, `../../REPORT.pdf`).